

2018 H2 MATH (9758/02) JC 2 PRELIM
SUGGESTED SOLUTIONS

Qn	Solution
1	Complex No.
(a)	If w is a complex root to the equation $P(z) = 0$, then $P(w) = 0$. $a_0 + a_1 w + a_2 w^2 = 0$ Taking conjugate on both sides of the equation (*): $(a_0 + a_1 w + a_2 w^2)^* = 0$ $(a_0)^* + (a_1 w)^* + (a_2 w^2)^* = 0$ Since $a_0, a_1, a_2 \in \mathbb{R}$, $a_0 + a_1 w^* + a_2 (w^*)^2 = 0$ $P(w^*) = 0$ Therefore, its complex conjugate w^* is also a root of $P(z) = 0$. <u>Alternative method</u> Let $w = x + yi$ where $x, y \in \mathbb{R}$ then $w^* = x - yi$. $a_0 + a_1 w + a_2 w^2 = 0$ $a_0 + a_1 (x + yi) + a_2 (x + yi)^2 = 0$ $a_0 + a_1 (x + yi) + a_2 (x^2 + 2xyi - y^2) = 0$ Comparing real and imaginary parts: $a_0 + a_1 x + a_2 (x^2 - y^2) = 0$ $a_1 y + a_2 (2xy) = 0$

	Subst $w^* = x - yi$ into $P(z)$: $a_0 + a_1 w^* + a_2 (w^*)^2$ $= a_0 + a_1 (x - yi) + a_2 (x - yi)^2$ $= a_0 + a_1 (x - yi) + a_2 (x^2 - 2xyi - y^2)$ $= a_0 + a_1 x + a_2 (x^2 - y^2) - [a_1 y + a_2 (2xy)]i$ Since $a_0 + a_1 x + a_2 (x^2 - y^2) = 0$ and $a_1 y + a_2 (2xy) = 0$ $a_0 + a_1 w^* + a_2 (w^*)^2$ $= 0 - 0i$ $= 0$ $\therefore P(w^*) = 0$
(b)(i)	$z_2 = -a - ai$
(b)(ii)	
(b)(iii)	Area of $OZ_1Z_2 = \frac{1}{2} \times (2a) \times a$ $= a^2$

Qn	Solution
2	Sequences & Series
(i)	$\text{LHS} = \cos\left(r + \frac{1}{2}\right)\theta - \cos\left(r - \frac{1}{2}\right)\theta$ $= \left[\cos r \theta \cos \frac{1}{2}\theta - \sin r \theta \sin \frac{1}{2}\theta \right] - \left[\cos r \theta \cos \frac{1}{2}\theta + \sin r \theta \sin \frac{1}{2}\theta \right]$ $= -2 \sin r \theta \sin \frac{1}{2}\theta = \text{RHS} \quad (\text{shown})$
(ii)	$\sum_{r=1}^n \sin r\theta$ $= \sum_{r=1}^n \frac{\cos\left(r + \frac{1}{2}\right)\theta - \cos\left(r - \frac{1}{2}\right)\theta}{-2 \sin \frac{1}{2}\theta}$ $= \frac{1}{-2 \sin \frac{1}{2}\theta} \sum_{r=1}^n \left[\cos\left(r + \frac{1}{2}\right)\theta - \cos\left(r - \frac{1}{2}\right)\theta \right]$ $= \frac{1}{-2 \sin \frac{1}{2}\theta} \left[\cancel{\cos \frac{3}{2}\theta} - \cos \frac{1}{2}\theta \right.$ $\quad \left. + \cancel{\cos \frac{5}{2}\theta} - \cancel{\cos \frac{3}{2}\theta} \right.$ $\quad \left. + \cancel{\cos \frac{7}{2}\theta} - \cancel{\cos \frac{5}{2}\theta} \right.$ $\quad \left. + \dots \right.$ $\quad \left. + \cancel{\cos\left(n - \frac{1}{2}\right)\theta} - \cancel{\cos\left(n - \frac{3}{2}\right)\theta} \right.$ $\quad \left. + \cos\left(n + \frac{1}{2}\right)\theta - \cos\left(n - \frac{1}{2}\right)\theta \right]$ $= \frac{1}{-2 \sin \frac{1}{2}\theta} \left[\cos\left(n + \frac{1}{2}\right)\theta - \cos \frac{1}{2}\theta \right]$ $= -\frac{1}{2} \operatorname{cosec} \frac{1}{2}\theta \left[\cos\left(n + \frac{1}{2}\right)\theta - \cos \frac{1}{2}\theta \right]$
(iii)	$\sum_{r=5}^{62} \sin(r-2)\theta$ $= \sum_{r=3}^{60} \sin r\theta \quad (\text{replace } r \text{ by } r+2)$ $= \sum_{r=1}^{60} \sin r\theta - \sin \theta - \sin 2\theta$ $= \frac{1}{-2 \sin \frac{1}{2}\theta} \left[\cos\left(60 + \frac{1}{2}\right)\theta - \cos \frac{1}{2}\theta \right] - \sin \theta - \sin 2\theta$ When $\theta = \frac{\pi}{3}$,

$\sum_{r=5}^{62} \sin \left[(r-2) \left(\frac{\pi}{3} \right) \right] = \frac{1}{-2 \sin \frac{\pi}{6}} \left[\cos \left(20\pi + \frac{\pi}{6} \right) - \cos \frac{\pi}{6} \right] - \sin \frac{\pi}{3} - \sin \frac{2\pi}{3}$ $= -1 \left[\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right] - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = -\sqrt{3}$

Qn	Solutions
3	Applications of Differentiation: Rate of Change, Maxima/Minima
(a)	$P = 2r + \pi r \quad \Rightarrow \frac{dP}{dr} = 2 + \pi$ $A = \frac{\pi r^2}{2} \quad \Rightarrow \frac{dA}{dr} = \pi r$ $\therefore \frac{dP}{dA} = \frac{\frac{dP}{dr}}{\frac{dA}{dr}} = \frac{2 + \pi}{\pi r} \quad (\text{Shown})$
	Given that $\frac{dP}{dt} = \frac{3}{5} \text{ cm/s}$ and $\frac{dA}{dt} = 3 \text{ cm}^2/\text{s}$ $\frac{dP}{dA} = \left(\frac{dP}{dt} \right) \left(\frac{dt}{dA} \right)$ $\frac{2 + \pi}{\pi r} = \left(\frac{3}{5} \right) \left(\frac{1}{3} \right)$ $5(2 + \pi) = \pi r$ $r = \frac{5(2 + \pi)}{\pi} \text{ cm}$
(b)	Let S be the external surface area of the model. $S = 4 \left(\frac{1}{2} \right) (2r) (\sqrt{2}r) + 4(2r)h + (2r)(2r)$ $= 4(\sqrt{2})r^2 + 8rh + 4r^2$ $= 4(\sqrt{2} + 1)r^2 + 8rh = 160$ $h = \frac{160 - 4(\sqrt{2} + 1)r^2}{8r}$ $= \frac{40 - (\sqrt{2} + 1)r^2}{2r}$ Volume of model $V = \frac{1}{3}(2r)(2r)r + (2r)(2r)h$ $= \frac{4}{3}r^3 + 4r^2 \left(\frac{40 - (\sqrt{2} + 1)r^2}{2r} \right)$ $= \frac{4}{3}r^3 + 2r(40 - (\sqrt{2} + 1)r^2)$ $= \frac{4}{3}r^3 + 80r - 2(\sqrt{2} + 1)r^3$ $= 80r - \left(\frac{2}{3} + 2\sqrt{2} \right) r^3$ For V to be maximum,

$$\frac{dV}{dr} = 80 - 3\left(\frac{2}{3} + 2\sqrt{2}\right)r^2 = 0$$

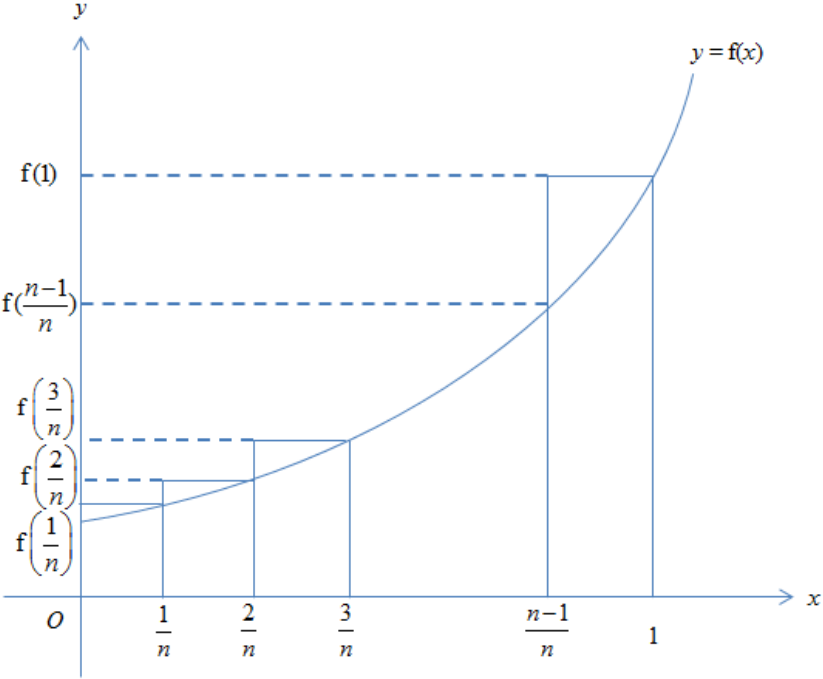
$$(4 - 6(\sqrt{2} + 1))r^2 + 80 = 0$$

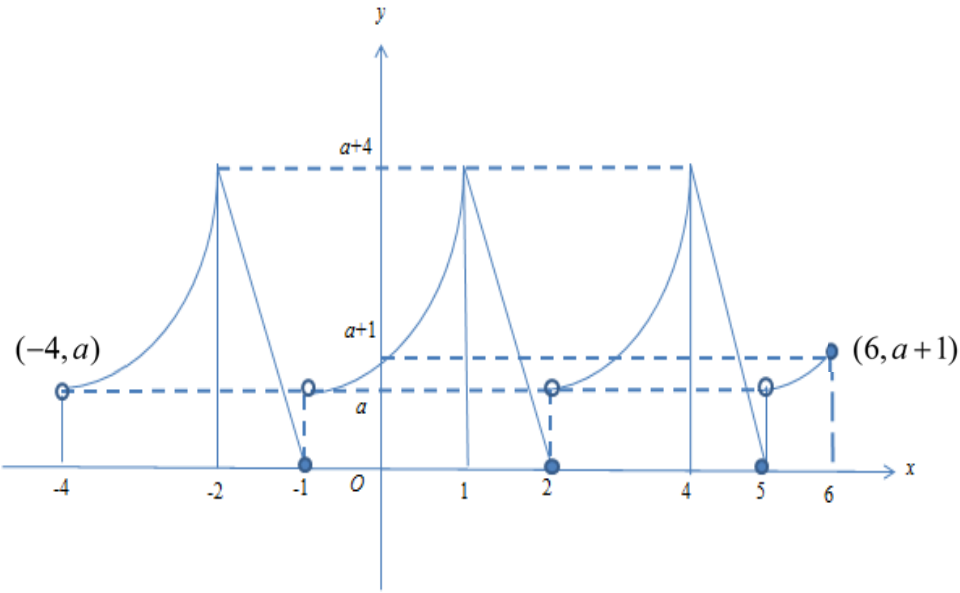
Since $r > 0$,

$$r = \sqrt{\frac{-80}{4 - 6(\sqrt{2} + 1)}}$$

$$= 2.7622$$

$$\approx 2.76 \text{ cm}$$

Qn	Solution
4	Applications of Integration
(a) (i)	 $\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right\}$ $= \lim_{n \rightarrow \infty} (\text{Total area of the } n \text{ rectangles})$ $= \text{Area under the curve } y = f(x) \text{ from } x = 0 \text{ to } x = 1$ $= \int_0^1 f(x) dx$
(a) (ii)	Let $f(x) = e^{3x}$ $\lim_{n \rightarrow \infty} \frac{1}{n} \left\{ e^{\frac{3}{n}} + e^{\frac{6}{n}} + \dots + e^3 \right\}$ $= \int_0^1 e^{3x} dx$ $= \frac{1}{3} [e^{3x}]_0^1$ $= \frac{1}{3} (e^3 - 1)$
(b)(i)	$f(-41) = f(1) = 4 + a$ $f(2018) = f(2) = 0$

(ii)	
(iii)	$\int_{-4}^4 f(x) dx$ $= 3 \int_{-1}^1 ((x+1)^2 + a) dx + 2 \left(\frac{1}{2} (1)(4+a) \right)$ $= \left[(x+1)^3 + 3ax \right]_{-1}^1 + 4 + a$ $= (8 + 3a) - (0 - 3a) + 4 + a$ $= 12 + 7a$

Qn	Solution
5	Permutations and Combinations
(i)	Number of ways = $8! = 40320$
(ii)	Number of ways = ${}^8C_1 \times 1 \times {}^6C_1 \times 1 \times {}^4C_1 \times 1 \times {}^2C_1 \times 1 = 384$ Alternative: $4! \times (2!)^4 = 384$
(iii)	Number of ways = $384 - {}^6C_1 \times 2 \times 4 \times 1 \times 2 \times 1 = 288$

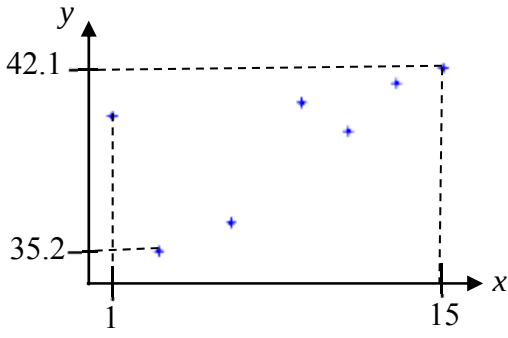
Qn	Suggested Solutions																																																
6	D.R.V.																																																
(i)	<table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>1</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>2</td><td>1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>3</td><td>2</td><td>1</td><td>0</td><td>1</td><td>2</td></tr><tr><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td><td>1</td></tr><tr><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr></table> Probability Distribution of X <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P(X = x)</td><td>$\frac{5}{25} = \frac{1}{5}$</td><td>$\frac{8}{25}$</td><td>$\frac{6}{25}$</td><td>$\frac{4}{25}$</td><td>$\frac{2}{25}$</td></tr></table>	X	1	2	3	4	5	1	0	1	2	3	4	2	1	0	1	2	3	3	2	1	0	1	2	4	3	2	1	0	1	5	4	3	2	1	0	x	0	1	2	3	4	P(X = x)	$\frac{5}{25} = \frac{1}{5}$	$\frac{8}{25}$	$\frac{6}{25}$	$\frac{4}{25}$	$\frac{2}{25}$
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(ii)	Using GC, $E(X) = 1.6$, $\text{Var}(X) = 1.44$ OR $E(X) = 0\left(\frac{1}{5}\right) + 1\left(\frac{8}{25}\right) + 2\left(\frac{6}{25}\right) + 3\left(\frac{4}{25}\right) + 4\left(\frac{2}{25}\right) = 1.6$ $E(X^2) = 0^2\left(\frac{1}{5}\right) + 1^2\left(\frac{8}{25}\right) + 2^2\left(\frac{6}{25}\right) + 3^2\left(\frac{4}{25}\right) + 4^2\left(\frac{2}{25}\right) = 4$ $\text{Var}(X) = E(X^2) - [E(X)]^2 = 4 - 1.6^2 = 1.44$																																																
(iii)	$E(2X - a) < 0$ $2(1.6) - a < 0$ $\therefore a > 3.2$																																																

Qn	Suggested Solutions
7	Binomial Distribution
(i)	(I) Whether a patient has diabetes is independent of any other patient who has diabetes. (S) The probability that a patient has diabetes is constant for all patients.
(ii)	(I) Patients who are selected may be family members hence each of them having diabetes are not independent of each other. OR (S) The probability of a patient having diabetes is not a constant as the elderly have a higher probability of having diabetes. OR any reasonable factors e.g. lifestyle, diet, race, age
(iii)	Let X be the number of patients, out of 24, who have diabetes. $X \sim B\left(24, \frac{1}{9}\right)$ Probability that in a sample of 25, the 25th patient is the third patient who has diabetes $= P(X = 2) P(\text{the 25th patient is the third patient who has diabetes})$ $= 0.25531 \times \frac{1}{9} = 0.0284$
(iv)	$P(X \leq n) > 0.9$ Using GC, When $n = 4$, $P(X \leq 4) = 0.87974 < 0.9$ When $n = 5$, $P(X \leq 5) = 0.95653 > 0.9$ Therefore least $n = 5$

Qn	Solution
8	Probability (Tree Diagram)
(i)	The tree diagram shows the following paths and probabilities: 1st Time: B ($\frac{5}{10}$), R ($\frac{2}{10}$), G ($\frac{3}{10}$) 2nd Time (from B): B ($\frac{5}{10}$), R ($\frac{2}{10}$), G ($\frac{3}{10}$) 3rd Time (from BB): B ($\frac{5}{10}$), R ($\frac{2}{10}$), G ($\frac{3}{10}$) 3rd Time (from BR): B ($\frac{5}{10}$), R ($\frac{2}{10}$), G ($\frac{3}{10}$)
(ii)	P(player wins \$12 or more when the game ends) $= P(BRR) + P(BBR) + P(R)$ $= \frac{5}{10} \left(\frac{2}{10} \right) \left(\frac{2}{10} \right) + \frac{5}{10} \left(\frac{5}{10} \right) \left(\frac{2}{10} \right) + \left(\frac{2}{10} \right)$ $= \frac{27}{100}$
(iii)	P (second spin lands on a blue sector the player wins \$12 or more when the game ends) $= \frac{\left(\frac{5}{10} \right) \left(\frac{5}{10} \right) \left(\frac{2}{10} \right)}{0.27} = \frac{5}{27}$
(iv)	P (no spins result on a red sector and wins nothing when game ends) $= \frac{3}{10} + \left(\frac{5}{10} \right) \left(\frac{3}{10} \right) + \left(\frac{5}{10} \right)^2 \left(\frac{3}{10} \right) + \left(\frac{5}{10} \right)^3 \left(\frac{3}{10} \right) + \dots$ $= \frac{3}{10}$ $= \frac{3}{1 - \frac{5}{10}}$ $= \frac{3}{5}$

Qn	Solution
9	Hypothesis Testing
(i)	Let X be the duration a randomly chosen shopper spends at MJ mall per visit (in hours). Let μ denote the population mean duration spent by shoppers at MJ mall per visit (in hours). $H_0: \mu = 1.5$ $H_1: \mu > 1.5$ $\bar{x} = \frac{\sum x}{n} = \frac{82.5}{50} = 1.65$ $s^2 = \frac{50}{49}(0.425) = 0.43367 \text{ (to 5 s.f.)} = 0.434 \text{ (to 3 s.f.)}$
(ii)	Since sample size is large, by Central Limit Theorem, the sample mean duration spent by 50 shoppers is approximately normal. $\bar{X} \sim N\left(1.5, \frac{0.43367}{50}\right)$ approximately. Hence there is no need to assume that the distribution of the duration shoppers spend at MJ mall per visit is normal.
(iii)	Using GC, Under H_0, $p\text{-value} = 0.053630$ (to 5 s.f.) $= 0.0536$ (to 3 s.f.) 0.0536 is the probability of obtaining a sample mean more than or equal to 1.65, assuming that the mean duration spent by shoppers at MJ mall is 1.5 hours. For management's hypothesis to be valid, H_0 is to be rejected. $p\text{-value} = 0.053630 < \frac{\alpha}{100}$ $\Rightarrow \alpha > 5.3630$ $\therefore$ Required set of values is $\{\alpha \in \mathbb{R} : 5.36 < \alpha \leq 100\}$ (to 3 s.f.)
(iv)	$H_0: \mu = 150$ $H_1: \mu < 150$ Given: $p\text{-value} < 0.05$ (a) This is necessarily true because $p\text{-value} < 0.05 < 0.1$ (b) This is not necessarily true nor necessarily false. Now, $H_0: \mu = 150$ $H_1: \mu \neq 150$ The new p-value is double of the original p-value. Case 1: If the original p-value was more than 0.025, for example 0.03, then the new p-value would be $0.06 > 0.05$, hence in such a case there would not be significant evidence at the 5% level of significance that the population mean spending differs from \$150.

	Case 2: If the original p-value was less than 0.025, for example 0.02, then the new p-value would be $0.04 < 0.05$, hence in such a case there would be significant evidence at the 5% level of significance that the population mean spending differs from \$150.
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Qn	Solution
10	Correlation & Relation
(i)	
(ii)	The outlier is (1, 40.3). Any appropriate reason, some possible reasons: • A fire in the dry grassland killed the wild rabbits • Drought/Famine in the region causes wild rabbits to die • Disease/Epidermic in the region killed the wild rabbits • Introduction of a new predator like coyote or dogs that hunts the wild rabbits, thus leading to a sudden decrease in the population. • Hunting season lead to a sudden decrease in the wild rabbit population.
(iii)	The population of wild rabbits <u>cannot increase or decrease indefinitely</u> in a region, hence a linear model is not appropriate or the population of wild rabbits should plateau due to limited resources.
(iv)	Using G.C., When $M = 45$, $r = -0.946\ 203$ (to 6 decimal places)
(v)	Since $M = 45$ gives a value of r that is closer to -1 , it would be the most appropriate value for M .
(vi)	Using G.C., $\ln(45 - y) = 2.6356 - 0.10435x$ $\Rightarrow \ln(45 - y) = 2.64 - 0.104x \quad (3 \text{ s.f.})$ $\Rightarrow a = 2.64, \quad b = -0.104$ When $x = 24$, $y = 45 - e^{2.6356 - 0.10435(24)} = 43.8598$ The population of the wild rabbits after 2 years is 4386.
(vii)	$M = 45$ i.e. 4500 rabbits is the <u>maximum population</u> of wild rabbits which this region can support in the long run, assuming all factors stay the same.

Qn	Solution
11	Normal Distribution & Sampling
(a)	If $T \sim N(5, 8)$, then $P(T < 0) = 0.0385$, which means that there is a non-zero probability that the time taken could be negative. However, this is impossible. Hence it is not suitable. OR If $T \sim N(5, 8)$, then 99.7% of the values would lie within $5 \pm 3\sqrt{8}$, which contains a significant range of negative values.
(b) (i)	Let X and Y be the masses of a randomly chosen chicken and a randomly chosen duck in kilograms respectively. $X : N(2.4, 0.5^2)$ $Y : N(4.3, 1.8^2)$ $X_1 + X_2 : N(4.8, 0.5)$ $P(X_1 + X_2 > 5) = 0.38865 \approx 0.389$ (3s.f.)
(b) (ii)	$\frac{X_1 + X_2 + Y}{3} : N\left(\frac{91}{30}, \frac{187}{450}\right)$ $P\left(\frac{X_1 + X_2 + Y}{3} > 3.2\right) = P\left(\frac{X_1 + X_2 + Y}{3} > 3.2\right)$ $= 0.39799 \approx 0.398$ (3s.f.) <u>Alternatively (Consider sum)</u> $X_1 + X_2 + Y : N\left(9.1, \frac{187}{50}\right)$ $P\left(\frac{X_1 + X_2 + Y}{3} > 3.2\right) = P(X_1 + X_2 + Y > 9.6) = 0.39799 \approx 0.398$ (3s.f.)
(b) (iii)	$T = 0.9(7)(X_1 + X_2) + 0.75(13)Y : N(72.165, 327.8475)$ $P(T < 80) \approx 0.667$ (3s.f.)